OXIDO

OXIDO: A Company That Runs Itself

AI agents that operate the business — on a training layer institutions can actually audit

383 Controlled Experiments • AAAI 2027 Submitted • 12-Month Live Ablation

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One Decision in 1986. Three Problems Everyone Funds Separately.

A Single Global Error Signal Makes Fault Localization Mathematically Impossible — Every Downstream Failure Follows From It

NOTHING CAN BE LOCATED

One error signal spreads across every parameter at once.

No unit can assess its own state.

The loss curve collapses and nothing says which node died.

NOTHING IS RETAINED

New tasks overwrite prior representations indiscriminately.

ResNet-18 keeps 47.3% after a sequential task shift.

Every regime change costs what the model knew.

NOTHING IS REPAIRED

Repair requires knowing what is broken.

Corrupted data is learned silently and confidently.

Rollback or full retrain are the only recourse.

One cause: the chain rule multiplies every unit's contribution through every other unit's. Unchanged since 1986.

Kill a Converged Layer. One Architecture Comes Back.

Bounding Each Node's Local Gradient Chain Produces Attribution as a Physical Byproduct — and Attribution Makes Repair Possible

STANDARD BACKPROPAGATION

One global error signal, entangled across all parameters.

- No per-node visibility.
- Silent collapse on node death.
- No diagnosis, no targeted repair.

Feature layer zeroed at epoch 100: 10.0% — chance, every seed, zero variance, permanent.

ORMAS THREE-SIGNAL TRAINING

Each node reads through a 4,416-parameter bottleneck — too small to memorise noise.

- Scores its own health every step.
- Fires at 2.5σ below its own history.
- Diagnoses 7 modes, repairs mid-run.

Same attack: 80.3% after 85 targeted corrections (+70.3pp).

Local stability characterization from Input-to-State Stability (Sontag, 1989) — not a global convergence proof, not yet peer-reviewed.

The Industry Is Moving From Inference to Training on Private Data

Three Independent Pressures Converge on the Same Missing Layer — None of Them Existed Two Years Ago

REGULATORS SHIFTED TO ARCHITECTURE

The FDA rejects medical models for opacity regardless of accuracy.

The EU AI Act attaches liability to systems with no audit trail.

The requirement is no longer performance. It is provenance.

RENTED INFERENCE STOPPED SCALING

Per-call cost compounds linearly with usage.

At institutional volume the only exit is your own infrastructure.

Which means training on your own data — the thing nobody can do safely.

MODELS DEGRADE INVISIBLY

Every regime change, patient cohort and fraud pattern erodes what the model knew.

No detection exists at the training layer.

It surfaces as a miscalculation, not an alert.

The regime is shifting from inference on public data to training on proprietary institutional data.

Fifty Accounts Is a $10M Business. Here Is Who They Are.

Counted Upward From the Regulatory Constraint — Not Down From a Market Report

THE QUALIFYING BUYER

Three properties at once, or not a customer:

1. Holds data that legally cannot leave its infrastructure.
2. Has an active initiative to train on it.
3. Answers to a regulator that will ask what the model learned.

The constraint makes the count small and the contracts large.

CUSTOMERS #1–#50, BY SEGMENT

#1–10 Academic medical centres — ~180 with an internal AI group *and* a regulated pathway. The budget line exists before we arrive.

#11–30 Quant funds & asset managers — ~400 with in-house ML and model-risk duties.

#31–50 Insurers & credit risk — ~270 under continuous drift and supervisory reporting.

~850 accounts × $200K ACV → 50 = $10M ARR

Order-of-magnitude estimates. **No buyer interviews yet** *— the primary open risk, stated rather than assumed away.*

Three Layers. Each One Exists Because the Last One Broke.

An Operating System That Runs the Company, a Training Layer That Lets It Learn, a Model That Removes the Last Dependency

LAYER 1 — OXIMO

The operating system

Agents decompose an objective, design and validate the specialists they lack, and execute through a three-tier hierarchy.

Memory persists across sessions.

Ran a live company for 12 months, every operational role an agent.

LAYER 2 — ORMAS

The training layer

Built because Layer 1 could not learn from the data the business produced.

Per-node health, 7 diagnosed failure modes, bounded repair mid-run.

383 controlled experiments, 4 architectures, one consumer GPU.

LAYER 3 — THE MODEL

Not built yet — the endgame

Pre-trained on the three-signal architecture from scratch, not fine-tuned on someone else's.

Removes the external API dependency.

At that point the stack rents intelligence from no one.

anonymous.4open.science/r/ormas-EB73 · /oximo-5C73 · zenodo.org/records/21730363 — all 383 runs reproduce with one command.

I Registered a Company to Try to Disprove My Own Architecture

Seven Escalating Tests Over Twelve Months, Each Designed to Fail in Public — All Seven Held

RUNGS 1–4

1. Origination — demand from nothing? Cold start, no brand, no ad budget. Customers arrived, 100% LLM-referred.
2. Unit economics — $0.0043 per 12-asset suite vs $50–$150 human.
3. Causal isolation — was it the AI or the market? System stripped, all else identical. −91% output.
4. Collapse morphology — −100% new customers, all channels at once.

RUNGS 5–7

5. Reversibility — lesion or permanent? Re-injected, nothing else changed. Immediate recovery.
6. Recovery magnitude — +1,300%, overshooting 3.3× on accumulated memory alone.
7. Transaction ceiling — can it close something expensive? $4,386 order at $0.00 CAC, no human involved.

The scale stops there because I stopped it — no jurisdiction assigns liability to an autonomous agent.

The simultaneity is the proof, not the magnitude. Aug 2025–Aug 2026: 396 paying customers, 10 countries, £0 advertising.

Give Away the Operating System. License the Immune System.

Free Adoption Drives Distribution; the Training Layer Carries the Revenue and the Switching Cost

TIER 1 — FREE

The operating system ships open source.

It closes the gap between one person and a ten-person team, which is why it gets adopted.

Generates no revenue on purpose.

Content structured for machines gets recommended by machines.

TIER 2 — LICENSED

The training layer deploys inside the client's own infrastructure.

$150K–$300K per year, benchmarked against model-risk tooling.

Zero data leaves the perimeter, so infrastructure cost is near zero and margin is software-grade.

THE RENEWAL

Renewal is not satisfaction-driven.

After 12 months of accumulated agent memory, removal is not a migration.

That experiment has been run: taking the system out cost 91% of output within weeks.

Measured, not modelled.

Zero enterprise contracts signed to date. *The training layer had to be finished before it could be sold.*

The Regulator Creates the Demand. A Person Has to Close It.

Compliance Deadlines Generate Inbound Pull — the Remaining Gap Is Someone Who Can Sit in the Procurement Meeting

WHY THEY COME TO US

The buyer already has the problem written down by someone else.

- Regulators reject models on architecture, not accuracy — the requirement is documented before we arrive.
- The audit trail is produced during training, not reconstructed after.
- Nothing leaves the client's infrastructure — removing the objection that kills every SaaS alternative.

THE ONE GAP

The system replaced every operational role in a company, with causal data on how much of it was the system.

It cannot sit in a compliance review and be a person the buyer holds accountable.

Regulated institutions buy from someone who has survived their procurement process.

That is the co-founder. Not an engineer, not a generalist.

Design partner → single-model deployment → organisation-wide licence. Regulated cycles run 6–18 months.

Everyone Is Competing One Layer Above the Constraint

Orchestration Is Not the Blocker — the Blocker Is That You Cannot Safely Train on the Data

	Per-node attribution	Stability analysis	Autonomous repair	Training-time audit trail	Runs on their metal
Agent frameworks	×	×	×	×	×
Enterprise data platforms	×	×	×	∼	∼
Frontier model vendors	×	×	×	×	×
Data labelling vendors	×	×	×	×	×
Academic research	∼	∼	∼	×	—
This	✓	✓	✓	✓	✓

The moat is not secrecy — the work is published. It is that reaching this required breaking a live company on its own data first, then control theory, then two years of experiments. That order does not compress.

The assets worth owning are the deployment record, the accumulated institutional memory, and a model nobody else has started.

Everything Technical Is Built. One Thing Is Not.

Three Years Solo Produced the Architecture, the System and the Causal Proof — the Gap Is a Person, Not a Capability

FOUNDER

Eighteen. No university, no lab, no advisor, no external funding.

A self-correcting training architecture with a formal stability result, derived at 17.

An agent operating system that ran a live company for twelve months.

A UK entity registered from abroad at 17, unassisted, against an AML barrier.

Every experiment ran on one GPU, bought at 15 with money from the first thing he sold.

CO-FOUNDER SOUGHT

One specific profile:

Has sold software inside a regulated data environment — a fund, a hospital, an insurer.

Understands model risk well enough to argue with a compliance officer and win.

Not an engineer. Not a generalist. *Equal equity, four-year vest.*

The honest gap: never having worked inside a team. raadh.me · zenodo.org/records/21730363

From a Training Layer to the Substrate Regulated AI Runs On

Every Phase Removes One Dependency — Ending With a Stack That Rents Intelligence From No One

PHASE 1 — NOW

Finish the research and carry per-node health monitoring onto attention heads.

Architecture-agnostic by design, so this is a compute problem, not a research one.

First design partners.

PHASE 2 — 12–24 MONTHS

First licensed deployments inside institutions that cannot move their data.

Each accumulates institutional memory that cannot be exported.

Switching cost compounds with tenure.

PHASE 3 — YEARS 3–5

A model pre-trained on the self-correcting architecture from scratch.

The external dependency disappears; cost collapses to hardware.

A position no competitor reaches by adding a feature.

The network learns to stop needing correction. That is the thesis, and it is measured.

Correction frequency decays 4.2 → 0.05 per epoch — the system repairs itself until it no longer needs to.